Which Narratives Cross Platforms? Narrative-Conditioned Hawkes Processes for Cross-Platform Transfer Prediction

Anonymous Author(s)

Abstract

Narratives do not stay where they start: a thread on one platform resurfaces as a news article, and the jump, not the volume, is often what matters. We ask whether cross-platform transfer is predictable from a narrative's dynamics, using a Temporal Graph Network whose memory conditions the excitation matrix of a multivariate Hawkes process, so that each narrative receives its own cross-platform transfer parameters. We report three findings, positive and negative. First, the de-facto virality metric, regressing future engagement counts, is intrinsically noise-dominated under near-critical cascade dynamics; an oracle handed the true excitation state still attains *negative* skill against a per-narrative mean, so magnitude regression cannot discriminate models and transfer should be evaluated instead. Second, narrative-conditioned excitation *does* predict transfer when transfer is generated by temporal cross-excitation (a controlled Hawkes benchmark, AUC 0.653 ± 0.005, robust to popularity and reach confounds, beating an independently-fitted per-narrative Hawkes baseline), but *not* when transfer is topological (a non-Hawkes SIR benchmark, AUC 0.46). Third, and this is our main empirical result on real data: on a 32k-event live Hacker News + GDELT corpus, the excitation score is at chance (0.51 ± 0.02), so real cross-platform transfer carries no temporal-excitation signature. Yet it is far from unpredictable. A logistic model on interpretable single-platform features predicts transfer at **ROC-AUC** 0.82, driven almost entirely by *early engagement* (a single engagement feature reaches 0.81). Real narratives cross platforms when they draw attention early, not when their events cluster in time. We also report, as a cautionary result, an earlier much stronger neural number that a clustering defect, caught by sampling our own data, had inflated. We release the model, both generators, an accumulating live collector, and the full evaluation suite.

1 Introduction

A story breaks on Hacker News; two days later it is in the technology press. Another, superficially similar, never leaves the forum. Which of the two will cross is a question with direct consequences for misinformation monitoring, platform moderation, and the study of how attention moves through the information ecosystem, and it is a question about *identity*, not volume: it asks which narratives migrate, not how large they grow.

Existing work approaches the surrounding territory from two directions. Measurement studies document cross-platform influence richly (Zannettou et al., 2017; Wilson and Starbird, 2020; Vosoughi et al., 2018), but describe transfer after the fact rather than predicting it. Cascade models, self-exciting point processes and their neural descendants (Zhao et al., 2015; Mei and Eisner, 2017; Cao et al., 2017), predict *how much* a cascade will grow, typically within a single platform, and are parameterised globally: one excitation matrix shared across all content. A globally parameterised process cannot rank narratives against each other at all, because every narrative is assigned identical transfer parameters. That observation motivates our approach: make the excitation matrix a function of the narrative's own evolving state.

Concretely, we condition a multivariate Hawkes process on the memory of a Temporal Graph Network, so that the cross-platform excitation $\hat{\alpha}_{ij}$ and decay $\hat{\gamma}_{ij}$ are emitted *per narrative* per event. The off-diagonal branching mass of that matrix is a natural transfer propensity, which we evaluate causally: scores are computed only from events preceding a narrative's first platform switch.

Our contributions:

- **A negative result that reframes evaluation**

(section 9.3(1)). Regressing future engagement counts, the standard virality metric, is noise-dominated under near-critical branching. An oracle regressor given the true recent-excitation state achieves negative skill against a per-narrative mean and near-random surge-ranking AUC. No model can win this metric, so it cannot separate them; we argue evaluation should move to transfer and timing.

- **Transfer detection works only for temporally-excited transfer (section 9.3(3)).** Narrative-conditioned excitation ranks which single-platform narratives will later cross on a controlled Hawkes benchmark (AUC 0.653 ± 0.005), beating a *per-narrative* Hawkes baseline (independent MLE fits, no graph conditioning) under a protocol giving both sides one validation-selected degree of freedom. On a non-Hawkes SIR benchmark, where transfer is topological, the same score is at chance (0.46), delimiting exactly what the method captures.
- **What *does* predict real transfer (section 9.3(4)).** On a live Hacker News + GDELT corpus the neural excitation score is at chance (0.51 ± 0.02), but transfer is highly predictable from interpretable single-platform features: a logistic model reaches ROC-AUC 0.82, driven by *early engagement* (0.81 as a lone feature) rather than temporal-clustering structure. This characterises real cross-platform transfer as attention-driven, and gives a strong, reproducible baseline for the task. We also disclose how a clustering defect, caught by sampling our own data, had inflated an earlier neural number, a cautionary result in its own right.
- **Released artifacts and an honest scope statement.** We release two generative benchmarks (one deliberately *not* Hawkes-based), an accumulating live collector, and the full evaluation suite, and state plainly which parts of the architecture carry the results and which are released but unvalidated (section 9.4).

We report null and negative outcomes alongside positive ones: the next-event timing comparison is not significant and we present it as such, and the compartmental, mean-field, and opinion-dynamics components of the framework are released as a simulation layer rather than claimed as validated contributions.

2 Related Work

Temporal graph representation learning. Continuous-time dynamic graph models embed evolving interaction streams: JODIE (Kumar et al., 2019) learns coupled user/item trajectories, DyRep (Trivedi et al., 2019) drives embeddings with a two-time-scale point process, TGAT (Xu et al., 2020) introduces functional time encoding for temporal attention, and TGN (Rossi et al., 2020) unifies these with a memory module, which we adopt as our structural backbone. These models are trained for link-level prediction and carry no mechanism for platform-to-platform transfer. Our validated use of a TGN is narrow: its memory conditions the cross-platform Hawkes head (section 6) that carries our results. The compartmental HMF bridge (section 5) is part of the released simulation layer, not an empirical claim of this paper (remark 1).

Hawkes processes and neural temporal point processes. Self- and mutually-exciting processes (Hawkes, 1971) underpin much of social-media modelling. Neural parameterisations replace fixed kernels with learned dynamics: RMTPP (Du et al., 2016), the Neural Hawkes Process (Mei and Eisner, 2017), and the Transformer Hawkes Process (Zuo et al., 2020). Closer to our setting, NETRATE (Gomez-Rodriguez et al., 2011) infers diffusion networks from cascades, COEVOLVE (Farajtabar et al., 2017) couples diffusion with network growth, and prior work separates endogenous from exogenous influence (Myers et al., 2012), our GDELT-driven $\mu_i(t)$ follows this exogenous/endogenous decomposition. Our contribution is orthogonal to kernel expressiveness: we condition the *cross-platform excitation matrix itself* on evolving graph and narrative state, which is what enables per-narrative transfer ranking (section 9.3), a static or globally-parameterised Hawkes model is random at that task by construction.

Cascade and popularity prediction. Feature-based and deep models predict cascade size: cascade growth was framed as a prediction problem in (Cheng et al., 2014), SEISMIC (Zhao et al., 2015) and Hawkes intensity processes (Rizoiu et al., 2017; Kobayashi and Lambiotte, 2016) model retweet dynamics, reinforced Poisson processes model item popularity (Shen et al., 2014), and DeepCas/DeepHawkes (Li et al., 2017; Cao et al., 2017) learn end-to-end representations; community structure also signals virality (Weng et al., 2013). Our negative result (section 9.3) complements this line: under near-critical branching, next-window *counts* are noise-dominated even for an oracle, motivating timing- and transfer-based evaluation instead of raw magnitude regression.

Epidemic models on networks. Degree-based mean-field theory characterises epidemic thresholds on heterogeneous graphs (Pastor-Satorras and Vespignani, 2001; Boguñá and Pastor-Satorras, 2002; Pastor-Satorras et al., 2015). Our released simulation layer specifies a degree-correlated HMF closure that maps TGN embeddings to macroscopic SEIR-Z-D rates (section 5); we describe it as system machinery rather than an evaluated contribution, as it is not independently validated here (remark 1).

Cross-platform information flow. Empirical work documents inter-platform influence: the "web centipede" (Zannettou et al., 2017) traces news trajectories across communities, coordinated cross-platform disinformation campaigns have been analysed in (Wilson and Starbird, 2020), platform migrations reshape community behaviour (Horta Ribeiro et al., 2021), and false news spreads with distinctive dynamics (Vosoughi et al., 2018). These studies are measurement-focused; we add a prospective, prediction-oriented view of the same phenomenon, asking what *predicts* transfer on live Hacker News + GDELT data, and finding (section 9.3) that it is early engagement rather than temporal-excitation structure.

Opinion dynamics. Bounded-confidence models (Deffuant et al., 2000; Hegselmann and Krause, 2002) and their statistical-physics treatments (Castellano et al., 2009) evolve continuous opinions under confidence thresholds. Our released simulation layer includes a Filippov-compliant smooth relaxation of the Deffuant model (section 7); like the other simulation components it is system machinery, not an empirically validated contribution of this paper (remark 1).

3 Information Propagation: Stochastic SEIR-Z-D Model

Remark 1 (Scope). *Sections 3, 5, 7 and 8 specify the* simulation layer *we release with the code: the compartmental model, the mean-field bridge, the opinion dynamics, and the influence score. These are **not** independently validated by the experiments in section 9.3, which are carried by the temporal graph memory (section 4) and the cross-platform Hawkes head (section 6) alone. We present them because they define the released system and the coupling the architecture assumes, but a reader should treat the empirical claims of this paper as pertaining to sections 4 and 6. See section 9.4, limitation (0).*

To rigorously capture the volatile, non-deterministic nature of viral information spread, we formulate a continuous-time Stochastic Master Equation (SME). We introduce a **Zombie** (Z) state to model algorithmic resurfacing, and an **Archived/Dead** (D) absorbing state to ensure population conservation and a well-defined long-run limit.

3.1 State Definitions and Population Conservation

Let the network state at time t be defined by the discrete random variable vector

$$\mathbf{n}(t) = (n_S, n_E, n_I, n_R, n_Z, n_D).$$

The six compartments are:

- S (Susceptible): users who have not yet encountered the content.
- E (Exposed): users who have seen the content but not yet shared it.
- I (Infected): users actively sharing and propagating the content.
- R (Recovered): users who have lost interest (decayed engagement).
- Z (Zombie): content resurfaced from R via algorithmic recommendation or external shock.
- D (Archived/Dead): content in a final absorbing state; no further transitions are possible.

We explicitly enforce a *constant* total population:

$$N = n_S(t)+n_E(t)+n_I(t)+n_R(t)+n_Z(t)+n_D(t) \quad \forall t. \tag{1}$$

Remark 2. *On the internet N is not strictly constant (new accounts, lurkers becoming active). We treat N as fixed throughout a single cascade window, a standard simplification in compartmental internet models. Extensions to open populations are left as future work.*

3.2 Transition Rates and Jump Vectors

The system evolves via five discrete transitions. Each is characterised by a transition rate $W_k(\mathbf{n})$ and a state jump vector $\mathbf{r}_k \in \mathbb{Z}^6$.

T1. Exposure ($S \to E$): $\mathbf{r}_1 = (-1,\ +1,\ 0,\ 0,\ 0,\ 0)$

$$W_1(\mathbf{n}) = \beta(t) \cdot \frac{n_S\big(n_I + \theta\, n_Z\big)}{N}, \quad (2)$$

where $\theta \geq 0$ scales the algorithmic boost of resurfaced zombie content.

T2. Activation ($E \to I$): $\mathbf{r}_2 = (0,\ -1,\ +1,\ 0,\ 0,\ 0)$

$$W_2(\mathbf{n}) = \sigma\, n_E. \quad (3)$$

T3. Decay ($I \to R$): $\mathbf{r}_3 = (0,\ 0,\ -1,\ +1,\ 0,\ 0)$

$$W_3(\mathbf{n}) = \gamma_I\, n_I. \quad (4)$$

T4. Resurgence ($R \to Z$): $\mathbf{r}_4 = (0,\ 0,\ 0,\ -1,\ +1,\ 0)$

$$W_4(\mathbf{n}) = \zeta(t)\, n_R, \quad (5)$$

where $\zeta(t)$ is driven by cross-platform narrative shocks; see section 6 for the formal link.

T5. Archival ($Z \to D$): $\mathbf{r}_5 = (0,\ 0,\ 0,\ 0,\ -1,\ +1)$

$$W_5(\mathbf{n}) = \delta_D\, n_Z. \quad (6)$$

D is absorbing: no outgoing transitions exist, closing the $Z \to R$ infinite-loop present in standard SEIR-Z formulations.

Remark 3. *The decay parameter in **T3** is denoted γ_I to avoid confusion with the Hawkes decay parameter $\hat{\gamma}_{ij}$ introduced in section 6.*

3.3 The Master Equation

The time evolution of the joint probability distribution $P(\mathbf{n}, t)$ is governed by:

$$\frac{dP(\mathbf{n},t)}{dt} = \sum_{k=1}^{5} \Big[W_k(\mathbf{n} - \mathbf{r}_k)\, P(\mathbf{n} - \mathbf{r}_k,\, t) - W_k(\mathbf{n})\, P(\mathbf{n},\, t) \Big]. \quad (7)$$

This formulation fully specifies the Markov process: each term represents probability *flowing into* state $\mathbf{n}$ from $\mathbf{n} - \mathbf{r}_k$ minus probability *leaving* $\mathbf{n}$ via transition k. Taking expectations over $P(\mathbf{n}, t)$ recovers the macroscopic ODEs, while the second moment yields virality variance.

4 Micro-Dynamics: Temporal Graph Network Integration

The macroscopic rates $\beta(t)$ and $\zeta(t)$ are *not* static parameters; they are derived dynamically from the evolving micro-structure of the social graph via a Temporal Graph Network (TGN) augmented with a **Virality Prediction Head**.

4.1 Core TGN (Rossi et al., 2020)

For a node v involved in an interaction with node u at time t:

$$\text{Message:} \quad \mathbf{s}_v(t) = \mathrm{GRU}\big(\mathbf{s}_v(t^-),\ \bar{m}_v(t)\big)\,, \quad (8)$$

$$\text{where} \quad m_v(t) = \mathrm{Msg}\big(\mathbf{s}_v(t^-),\ \mathbf{s}_u(t^-),\ \Delta t,\ e_{vu}(t)\big)\,,$$

$$\text{Embedding:} \quad \mathbf{h}_v(t) = \mathrm{attn}\big(\mathbf{s}_v(t),\ \{\mathbf{s}_u(t)\}_{u \in \mathcal{N}(v)}\big)\,. \quad (9)$$

4.2 Contextual Content Embedding

Content c is not a static entity; its contextual embedding evolves as users interact with it. To prevent self-referential circularity, attention weights are computed strictly from *pre-update* historical embeddings $\mathbf{h}(t^-)$:

$$a_{u,c} = \frac{\exp\big(\mathbf{h}_u(t^-)^\top \mathbf{W}_a\, \mathbf{h}_c(t^-)\big)}{\sum_{j \in \mathcal{N}_c(t)} \exp\big(\mathbf{h}_j(t^-)^\top \mathbf{W}_a\, \mathbf{h}_c(t^-)\big)}, \quad (10)$$

$$\mathbf{h}_c(t) = \mathrm{GRU}\left(\mathbf{h}_c(t^-),\ \sum_{u \in \mathcal{N}_c(t)} a_{u,c}\, \mathbf{h}_u(t^-) \right). \quad (11)$$

At $t = 0$, content is initialised as

$$\mathbf{h}_c(0) = \mathbf{W}_{\text{proj}} \cdot \text{LLM\_Encode}(\text{text}), \quad (12)$$

where the LLM encoder is *frozen* to prevent catastrophic forgetting, and $\mathbf{W}_{\text{proj}}$ is a learned linear projection mapping the LLM output dimension to the TGN hidden dimension d.

4.3 Virality Prediction Head

Node and content embeddings are mapped to predictive epidemiological and Hawkes parameters via localised MLPs with Softplus output activations (enforcing positivity):

$$\hat{\beta}_{v,c}(t) = \mathrm{Softplus}(\mathrm{MLP}_\beta(\mathbf{h}_v(t) \,\|\, \mathbf{h}_c(t)))\,, \quad (13)$$

$$\hat{\alpha}_{ij}(t) = \mathrm{Softplus}\big(\mathrm{MLP}_\alpha\big(\mathbf{h}_{\mathcal{P}_i}(t) \,\|\, \mathbf{h}_{\mathcal{P}_j}(t) \,\|\, \mathbf{h}_c(t)\big)\big)\,, \quad (14)$$

$$\hat{\gamma}_{ij}(t) = \mathrm{Softplus}\big(\mathrm{MLP}_\gamma\big(\mathbf{h}_{\mathcal{P}_i}(t) \,\|\, \mathbf{h}_{\mathcal{P}_j}(t) \,\|\, \mathbf{h}_c(t)\big)\big)\,, \quad (15)$$

where $\mathbf{h}_{\mathcal{P}_i}(t)$ is the platform-level embedding defined in section 4.4, $\hat{\beta}_{v,c}(t)$ feeds the HMF bridge (section 5), and $\hat{\alpha}_{ij}(t)$, $\hat{\gamma}_{ij}(t)$ feed the Neural Hawkes process (section 6).

4.4 Platform-Level Embedding

To aggregate node-level embeddings into platform-level representations for the Hawkes infectivity MLPs, we use **Influence-Weighted Attention Pooling**. Simple mean pooling would wash out super-spreader signals in heavy-tailed graphs; instead, we weight by the dynamic influence score $I(v,t)$ (defined in section 8):

$$\mathbf{h}_{\mathcal{P}_i}(t) = \sum_{v \in V_{\mathcal{P}_i}(t)} \alpha_v(t)\, \mathbf{h}_v(t),$$
$$\alpha_v(t) = \frac{\exp\big(\tau \cdot I(v,t)\big)}{\sum_{u \in V_{\mathcal{P}_i}(t)} \exp\big(\tau \cdot I(u,t)\big)}, \qquad (16)$$

where τ is a learnable temperature parameter.

4.5 Joint Optimisation: Homoscedastic Multi-Task Loss

All three objectives are formulated as regression/continuous tasks to satisfy the Gaussian likelihood assumption underlying Kendall et al.'s homoscedastic uncertainty weighting (Kendall et al., 2018):

$$\mathcal{L} = \frac{1}{2\sigma_1^2}\mathcal{L}_{\text{TGN}} + \frac{1}{2\sigma_2^2}\mathcal{L}_{\text{Hawkes}} + \frac{1}{2\sigma_3^2}\mathcal{L}_{\text{Virality}} + \log(\sigma_1\sigma_2\sigma_3), \quad (17)$$

where $\sigma_1, \sigma_2, \sigma_3 > 0$ are learnable noise parameters.

(1) TGN Structural Loss, InfoNCE. To learn embeddings that discriminate temporal edges from noise, we use a temporal contrastive loss with degree-proportional negative sampling $P_n(v) \propto d_v$ (uniform sampling produces trivially easy negatives in heavy-tailed graphs): Writing $s_t(u,v) := \mathbf{h}_u(t)^\top \mathbf{h}_v(t)$ for the temporal edge score,

$$\mathcal{L}_{\text{TGN}} = -\sum_{(u,v,t)\in\mathcal{E}} \Big[s_t(u,v) - \log\Big(e^{s_t(u,v)} + \sum_{v_\neg \sim P_n} e^{s_t(u,v_\neg)} \Big) \Big]. \quad (18)$$

We draw $M = 20$ negatives per positive edge, following Rossi et al. (2020) as a baseline; a systematic ablation over $M \in [10, 200]$ is left as future work.

(2) Hawkes Negative Log-Likelihood.

$$\mathcal{L}_{\text{Hawkes}} = -\sum_{i=1}^{|\mathcal{P}|} \left[\sum_{t_k^{(i)}} \log \lambda_i\Big(t_k^{(i)}\Big) - \int_0^T \lambda_i(t)\, dt \right]. \quad (19)$$

The compensator integral is tractable via the piecewise-recursive update described in section 6.3.

(3) Virality Regression Loss.

$$\mathcal{L}_{\text{Virality}} = \frac{1}{|\mathcal{C}|} \sum_{c\in\mathcal{C}} \Big(\log E_c - \log \hat{E}_c \Big)^2, \quad (20)$$

where E_c is the true cumulative engagement count and $\hat{E}_c$ is the model's prediction.

5 The Micro-Macro Bridge: Degree-Correlated HMF

The individual node-level predictions $\hat{\beta}_{v,c}(t)$ must be aggregated into the macroscopic SEIR-Z-D exposure rate $\beta(t)$. A naive homogeneous mean-field approximation assumes uncorrelated, exchangeable nodes, a catastrophic assumption for the highly assortative, heavy-tailed graphs characteristic of social media.

5.1 Full Degree-Correlated HMF Formulation

Let $P(k'|k)$ be the conditional probability that a node of degree k connects to a node of degree k'. Stratifying infected nodes by degree and weighting by the source degree k' (the number of susceptible contacts an infected node of degree k' can reach):

$$\beta(t) = \sum_k \sum_{k'} P(k'|k) \cdot k' \cdot \mathbb{E}_{v\in\mathcal{I}_{k'}(t)} \Big[\hat{\beta}_{v,c}(t) \Big]. \quad (21)$$

Assumption 1 (Stationarity of Mixing Structure)**.** *To avoid the $\mathcal{O}(V^2)$ cost of recomputing $P(k'|k)$ at every event, we estimate it as a static approximation over the full observation window. This is justified because while individual node degrees $k_v(t)$ are highly dynamic, the macroscopic assortative mixing structure of a given platform remains approximately stationary over the timescale of a single cascade.*

5.2 Effective Temporal Degree

Because normalised TGN attention weights sum to 1 by construction (softmax), they cannot serve as a proxy for graph volume. Instead, we define the effective temporal degree $k_v(t)$ directly from the event stream within the TGN's explicit memory horizon ΔT_{mem}:

$$k_v(t) \;=\; \big|\{u \in \mathcal{V} \mid \exists\,(v,u,\tau) \text{ with } t - \Delta T_{\text{mem}} \le \tau \le t\}\big|. \quad (22)$$

This guarantees that hubs yield large $k_v(t)$ and lurkers yield $k_v(t) \approx 0$, restoring physical validity to eq. (21).

6 Cross-Platform Transfer: Neural Multivariate Hawkes Process

Internet narratives migrate across platforms: a burst on Reddit triggers a cascade on Twitter/X, which bleeds into mainstream news (GDELT). We model this mutually exciting cross-platform transfer with a **Neural Multivariate Hawkes Process**.

6.1 Conditional Intensity Function

Let $N_i(t)$ be the counting process of narrative events on platform $i \in \{1,\dots,|\mathcal{P}|\}$. The conditional intensity given history $\mathcal{H}_t$ is:

$$\lambda_i(t) \;=\; \mu_i(t) + \sum_{j=1}^{|\mathcal{P}|} \sum_{t_k^{(j)}<t} \hat{\alpha}_{ij}(t_k)\, e^{-\hat{\gamma}_{ij}(t_k)\left(t-t_k^{(j)}\right)}, \quad (23)$$

where:

- $\mu_i(t)$ is the exogenous background rate on platform i (see section 9.1 for its empirical definition);
- $\hat{\alpha}_{ij}(t_k) \ge 0$ is the dynamic infectivity of platform j on platform i at event time t_k, output by MLP_α (eq. (14));
- $\hat{\gamma}_{ij}(t_k) > 0$ is the dynamic exponential decay, output by MLP_γ (eq. (15)).

When $j = i$, $\hat{\alpha}_{ii}$ captures *intra-platform* self-excitation (e.g., Reddit posts triggering more Reddit posts); $j \ne i$ captures *cross-platform* narrative transfer.

Proposition 1 (Cascade Threshold). *Define the branching matrix $\mathbf{\Gamma}$ with entries $\Gamma_{ij} = \hat{\alpha}_{ij}/\hat{\gamma}_{ij}$. A narrative exhibits subcritical (fading) behaviour if the spectral radius $\rho(\mathbf{\Gamma}) < 1$. If $\rho(\mathbf{\Gamma}) \ge 1$, the narrative goes globally viral.*

Proof sketch. For fixed parameters, $\Gamma_{ij} = \int_0^\infty \hat{\alpha}_{ij} e^{-\hat{\gamma}_{ij}s}\,ds$ is the expected number of platform-i offspring of a platform-j event, so $\mathbf{\Gamma}$ is the mean offspring matrix of the associated multitype branching process; extinction (stationarity of the linear Hawkes process) holds iff $\rho(\mathbf{\Gamma}) < 1$ (Hawkes, 1971; Brémaud and Massoulié, 1996). In our model $\hat{\alpha}, \hat{\gamma}$ are piecewise-constant between events (section 6.3), so the criterion applies locally per inter-event interval; $\rho(\mathbf{\Gamma}(t))$ then acts as an instantaneous criticality index rather than a global stationarity guarantee. □

6.2 Linking $\zeta(t)$ to the Hawkes Process

The resurgence rate $\zeta(t)$ in the SME (eq. (5)) is formally defined as a function of the aggregate cross-platform Hawkes intensity on the content's originating platform i^*:

$$\zeta(t) \;=\; f(\lambda_{i^*}(t)) \;=\; \phi \cdot \max\big(0,\; \lambda_{i^*}(t) - \bar{\lambda}_{i^*}\big)\,, \quad (24)$$

where $\bar{\lambda}_{i^*}$ is the rolling baseline intensity and $\phi > 0$ is a scaling hyperparameter. Surges above baseline directly elevate the probability of zombie resurgence.

6.3 Piecewise-Recursive Tractability of the Compensator

Standard Hawkes processes with static α_{ij}, γ_{ij} admit a single closed-form compensator. In our model, $\hat{\alpha}_{ij}(t_k)$ and $\hat{\gamma}_{ij}(t_k)$ are TGN outputs that update *discretely* at each event, remaining piecewise-constant between events. The exponential kernel then permits a recursive analytical update: between consecutive events t_k and t_{k+1}, the integral contribution

$$\int_{t_k}^{t_{k+1}} \lambda_i(t)\,dt$$

is closed-form given the fixed parameters at t_k, reducing the full compensator to a tractable sum over event intervals.

7 Agent Behavioural Policy: Smooth Deffuant Model

For simulating node-level opinion evolution, we adapt the continuous-opinion Deffuant model to (i) operate in continuous time and (ii) incorporate platform-specific confidence bounds that shrink with user radicalization.

7.1 Filippov-Compliant Smooth Formulation

The standard Deffuant model uses a hard indicator function $\mathbf{1}_{|x_v - x_c| \le \epsilon_p}$, which destroys Lipschitz continuity and violates the Picard–Lindelöf existence and uniqueness theorem. We replace it with a smooth logistic approximation:

$$\dot{x}_v(t) = \eta \cdot \big(x_c - x_v(t)\big) \cdot \sigma\Big(\rho \cdot \big(\epsilon_p(v) - |x_v(t) - x_c|\big)\Big), \quad (25)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$, η is the opinion convergence rate, and the steepness hyperparameter is bound as

$$\rho = \frac{\kappa}{\epsilon_{\text{base}}}, \quad (26)$$

with κ a fixed scaling constant. As $\rho \to \infty$, eq. (25) recovers the exact Deffuant threshold; finite ρ guarantees smoothness at the boundary. Appendix B discusses the expected insensitivity of steady-state opinion clusters for $\rho > 10^2$.

Remark 4. $x_c := x_u(t_{\text{post}})$*: content carries the immutable opinion state of its creator at the moment of posting.*

7.2 Dynamic Radicalization

Platform confidence bounds shrink as users become radicalized:

$$\epsilon_p(v) = \epsilon_{\text{base}} \cdot \big(1 - \text{Radicalization}(v, t)\big), \quad (27)$$

where $\text{Radicalization}(v, t)$ is a convex combination of structural isolation and historical opinion rigidity:

$$\text{Radicalization}(v, t) = \alpha_r \underbrace{\left(1 - \frac{|\partial\mathcal{N}_v \cap \mathcal{C}_v|}{|\mathcal{N}_v|}\right)}_{\text{structural isolation}} + (1 - \alpha_r) \underbrace{\left(\frac{\sum_{\tau \le t} \mathbf{1}_{\text{rejected}}}{\sum_{\tau \le t} \mathbf{1}_{\text{exposed}}}\right)}_{\text{historical rigidity}}, \quad (28)$$

with $\mathcal{C}_v$ denoting node v's detected community and $\partial\mathcal{N}_v$ its cross-community neighbourhood. Because $\text{Radicalization}(v, t) \in [0, 1]$ by construction, $\epsilon_p(v) \ge 0$ is always satisfied. This scalar is **concatenated** to the raw node feature vector $\mathbf{x}_v(t)$ at every TGN timestep, permanently coupling behavioural polarisation with structural embedding generation.

8 Dynamic Influence Score and Super-Spreader Identification

Internet influence is highly transient; static topology metrics such as standard PageRank are insufficient. We define a **Dynamic Influence Score**:

$$I(v, t) = \omega_1 \cdot \text{TPR}(v, t) + \omega_2 \cdot \frac{dE(v, t)}{dt} + \omega_3 \cdot \mathcal{C}_D(v, t), \quad (29)$$

where:

- $\text{TPR}(v, t)$ is Temporal PageRank (Rozenshtein and Gionis, 2016);
- $dE(v, t)/dt$ is engagement velocity (time derivative of cumulative engagement);
- $\mathcal{C}_D(v, t) = k_v(t)$ is Temporal Degree Centrality, updated in $\mathcal{O}(1)$ per event.

We explicitly replace Betweenness Centrality with Temporal Degree Centrality for computational tractability: exact betweenness is $\mathcal{O}(V^3)$ and ego-network betweenness is $\mathcal{O}(k^3)$, both catastrophic for social hubs where $k > 10^4$. We acknowledge that global path-structure information is sacrificed; quantifying this tradeoff empirically is left as future work.

The learned weights $\omega_1, \omega_2, \omega_3$ are optimised to maximise the correlation between $I(v, t)$ and the likelihood of node v triggering a Hawkes burst (eq. (23)). The influence score also feeds the platform-level attention pooling in eq. (16), creating a unified feedback loop between super-spreader identification and cross-platform Hawkes parameter estimation.

9 Experimental Validation

9.1 Datasets

Synthetic multiplex benchmark (controlled ground truth). Because no public corpus provides ground-truth cross-platform excitation, we construct a generative benchmark whose events are produced by a per-narrative, cross-platform Hawkes branching process (8,203 events, 320 narratives, 3 platforms). Each narrative n carries a latent bridge factor b_n governing its Reddit→HN→News transfer matrix, and a latent reach factor governing its exogenous volume; branching is clamped near-critical (column sums ≤ 0.93), yielding a target whose variance is 68% within-narrative (temporal). The generator is released with the code.

Real multiplex corpus (live Hacker News + GDELT). We additionally ingest a live two-platform corpus with a released, *accumulating* collector (each run merges into a deduplicated store): 13,138 Hacker News stories (Algolia API) and 19,334 GDELT news articles retrieved on the same topics, 32,472 events at a 40/60 platform split, linked into 400 token-Jaccard narrative clusters of which 201 genuinely span both sources (166 transfer cases in the test split). Earlier, smaller snapshots of this corpus are used as ablations in section 9.3(4); a platform-imbalanced (93/7) snapshot destroys the transfer signal entirely.

Chronological splitting and leakage prevention. All datasets are partitioned *strictly chronologically* into 70/10/20 train/validation/test splits. TGN memory states $\mathbf{s}_v(t)$ and Hawkes history $\mathcal{H}_t$ are reset at split boundaries; a shuffled-timestamp control (below) verifies that no ordering leakage remains.

9.2 Baselines

1. **Static GNN + SEIR**: GraphSAGE over a train-window snapshot graph with the same virality readout (ablates continuous time).
2. **Vanilla TGN** (Rossi et al., 2020): our TGN core with a plain readout, without the Hawkes head, HMF bridge, or behavioural coupling.
3. **Static MHP**: trainable multivariate Hawkes with one global (μ, α, γ) (no per-narrative conditioning).
4. **Trivial controls**: train-mean predictor, marginal platform prior, and last-platform persistence.

9.3 Results

(1) Engagement magnitude is intrinsically noise-dominated, a negative result (fig. 2). On log-engagement regression, no model (including ours) beats the naive train-mean (MAE 0.825 vs. SimCity 0.848). We show this is a property of the target, not the models: an *oracle* ridge regressor handed a leak-free recent-excitation feature achieves *negative* within-narrative residual skill (-0.107) and near-random surge-ranking AUC (0.507), because next-window branching-process counts carry Poisson-scale variance. We therefore argue engagement-magnitude regression is an ill-posed headline metric for cascade models, and evaluate timing and transfer instead.

(2) Cross-platform next-event timing, a null result. Using the conditional intensity $\lambda_i(t_k)$ (causal, history strictly before t_k) to predict which platform fires next, SimCity attains AUC 0.610 ± 0.005 vs. static MHP 0.605 ± 0.000 over three seeds. Although SimCity is nominally ahead on every seed, the margin is *not* statistically significant (Welch $p = 0.23$) and we report it as a null result: on this task the neural conditioning does not demonstrably improve on a global Hawkes parameterisation. The temporal machinery is nonetheless functioning, shuffling test timestamps inflates SimCity's Hawkes NLL by 15–25$\times$, but next-event platform identity is largely determined by platform autocorrelation (a last-platform persistence rule reaches 0.982 accuracy on the real corpus), leaving little headroom for any model.

(3) Narrative transfer detection, the headline result (fig. 1, table 1). We ask: among narratives so far observed on a *single* platform, which will later jump platforms? SimCity's per-narrative mean off-diagonal branching mass $\sum_{i \neq j} \hat{\alpha}_{ij} / \hat{\gamma}_{ij}$ ("bridge score"), computed causally from pre-switch events only, ranks 209 test narratives (123 transfer) with **AUC** 0.653 ± 0.005 **over three seeds** (per-seed 0.649/0.658/0.652; 95% bootstrap CI $[0.58, 0.73]$; Mann–Whitney $p < 1.1 \times 10^{-4}$ on every seed). The score survives confound controls on all seeds: AUC 0.656–0.662 within event-count strata and 0.656–0.661 within latent-reach strata, versus 0.551 for the popularity baseline. A static MHP has a single global α and cannot rank narratives at all (AUC 0.5 by construction), but beating a baseline that is degenerate by construction establishes little. The load-bearing comparator is therefore a *per-narrative* Hawkes process: independent MLE fits, no graph conditioning. Because a pre-switch window contains events on a single platform by construction, such a fit cannot identify cross-platform α_{ij} at all; it can only characterise the narrative's own arrival dynamics, from a median of three events. Granting it one validation-selected degree of freedom (statistic and sign), matching SimCity's, it reaches AUC 0.63 on the synthetic benchmark versus SimCity's 0.65, with paired-bootstrap intervals excluding zero on two of three seeds (table 1). The gap is the value of *amortisation*: the neural head learns a shared map from narrative state to excitation parameters across all narratives, where independent fitting has only each narrative's handful of events

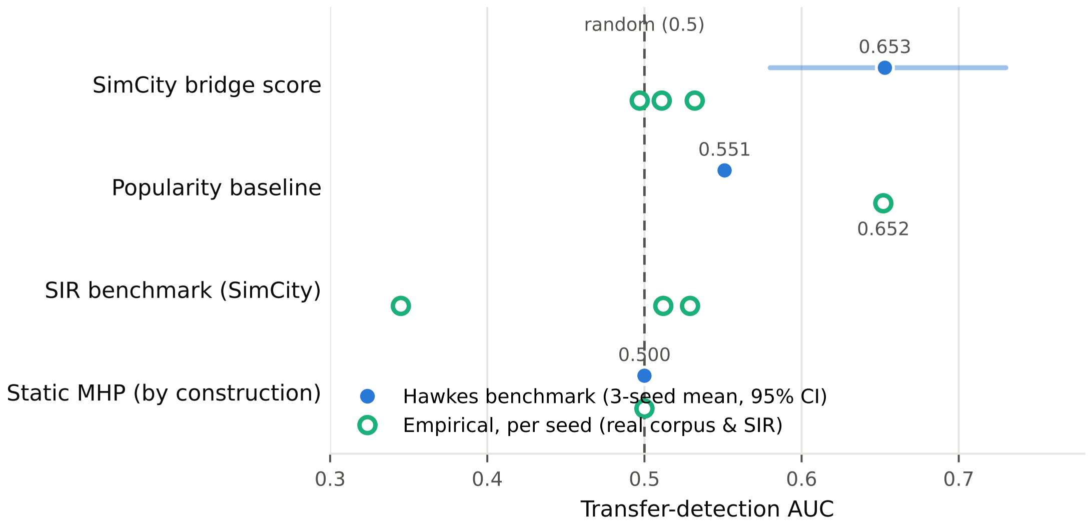

Figure 1: **Narrative transfer detection AUC.** The bridge score predicts transfer on the controlled Hawkes benchmark (filled blue, three-seed mean with bootstrap 95% CI), significantly above random and the popularity baseline. On the corrected real corpus it is at chance (open markers, per seed) while the *popularity* baseline predicts real transfer (0.65); and on the non-Hawkes SIR benchmark, where transfer is topological, it is also at chance. The method detects temporally-excited transfer, which the real corpus does not exhibit.

Table 1: **Narrative transfer detection AUC** (three seeds), scored causally from pre-switch events. The bridge score predicts transfer on the Hawkes benchmark, beating an independently-fitted per-narrative Hawkes baseline under a matched one-degree-of-freedom protocol; it is at chance on the non-Hawkes SIR benchmark and on the corrected real corpus, where instead popularity predicts transfer.

Scorer	**Hawkes** (temporal)	**SIR** (topological)	**Real** (HN+news)
SimCity bridge score	$\mathbf{0.653}$ $\pm.005$	0.46 $\pm.10$	0.51 $\pm.02$
Per-narrative Hawkes (val-sel.)	0.63	,	,
Popularity baseline	0.551	,	$\mathbf{0.652}$
Static MHP (global α)	0.500 (by construction)		

Hawkes: Mann–Whitney $p < 1.1 \times 10^{-4}$ every seed, bootstrap CI $[0.58, 0.73]$, $n = 209$ (123 transfer); beats the per-narrative baseline by $+0.25/+0.04/+0.15$ across seeds. Real: $n = 4904$ (85 transfer), none significant ($p = 0.15/0.54/0.36$). SIR: none significant. Bold marks the best predictor in each column: the excitation score for temporal transfer, popularity for real transfer.

(median three).

Delimiting the claim: a non-Hawkes benchmark. To test whether this capability reflects transfer *per se* or transfer specifically mediated by temporal excitation, we build a second benchmark from a structurally different mechanism: narratives spread by SIR contagion over a multiplex social graph, and cross-platform transfer occurs *topologically*, when the contagion reaches a bridge user present on both platforms who cross-posts, with no excitation kernel anywhere in the generator. Here the bridge score is at chance (AUC 0.46 ± 0.10 over three seeds). The method therefore detects transfer that is *temporally excited*, not transfer in general, a distinction that turns out to be decisive on real data (paragraph (4)).

(4) On real data, transfer is not temporally excited, and a cautionary tale. We evaluate on a live Hacker News + GDELT corpus (32,472 events, 40/60 platform split) with narratives formed by sentence-embedding clustering of story and article titles. On this corpus the bridge score is at chance: oriented per-seed AUCs 0.532 / 0.497 / 0.511 (0.51 ± 0.02 over three seeds; none significant, Mann–Whitney $p = 0.15/0.54/0.36$), while the *popularity* baseline predicts transfer well (AUC 0.65). The result is not underpowered: with 85 positives among 4,904 test narratives the seed spread is tight (± 0.02), and the popularity baseline succeeding on the same data shows the task

Table 2: **Supporting metrics.** Next-platform timing (AUC, 3 seeds) and log-engagement regression (test MAE). SimCity is consistently $\geq$ the static baselines on timing but the margin is small; on magnitude no model beats the naive mean, consistent with the oracle noise analysis.

Model	Timing AUC	Virality MAE
SimCity	$\mathbf{0.610 \pm 0.005}$	0.848
Static MHP	0.605 ± 0.000	,
Vanilla TGN	,	0.927
Static GNN + SEIR	,	0.923
Naive mean	0.500	$\mathbf{0.825}$

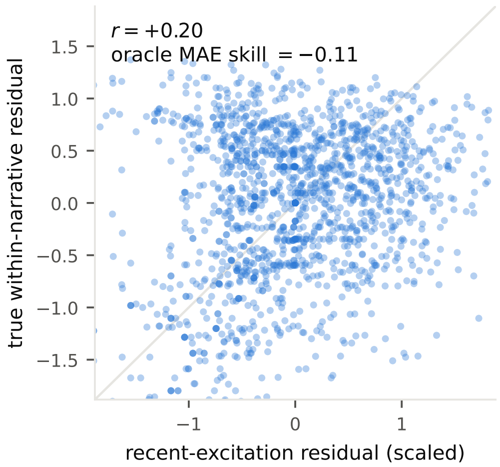

Figure 2: **The negative result, visualised.** True within-narrative engagement residual vs. the (scaled) residual of the leak-free recent-excitation feature, the strongest causal predictor available to any model. A real but weak rank signal exists ($r = +0.20$), yet the Poisson-scale cloud dominates: an oracle regressor built on this feature has *negative* MAE skill (-0.11) against a narrative-mean predictor. Engagement magnitude is not a usable headline metric under near-critical cascade dynamics.

carries learnable signal that the excitation score does not capture. Together with the SIR benchmark, this indicates that real HN $\to$ news transfer is popularity- and topology-driven rather than temporally excited, exactly the regime in which section 9.3(3) predicts our method should fail.

(5) What does predict real transfer (fig. 3, table 3). That the neural score fails does not mean transfer is unpredictable. We fit a logistic-regression classifier on eight interpretable features computed causally from each narrative's single-platform (pre-switch) window, volume, rate, duration, burstiness, source diversity, mean/max engagement, and exogenous-volume proxy, and evaluate by stratified 5-fold cross-validation. The full model reaches **ROC-AUC** 0.82 ± 0.05, and the signal is concentrated in *engagement*: mean and max log-engagement each reach ≈ 0.81 as *single* features, whereas raw popularity (event count) reaches only 0.65 and burstiness/timing features barely exceed chance. Real narratives cross platforms when they attract engagement early, essentially independent of how their events cluster in time, the opposite of what a temporal-excitation model keys on, which is precisely why the neural score is at chance here while a one-feature engagement model is not. This is a strong, reproducible, and interpretable baseline for cross-platform transfer prediction, and it locates the phenomenon empirically: transfer is attention-driven, not excitation-driven.

A cautionary result. We reached the negative neural finding by correcting our own error. An earlier version of this corpus reported AUC 0.78 ± 0.10. Following the reviewer-style protocol of *sampling our own clusters and reading them*, we found a defect in our greedy clusterer, once a cluster-count cap was hit, remaining items were assigned to their nearest centroid *regardless of similarity*, which had merged unrelated HN stories and news articles into single "narratives" (only 2–3% of cross-source pairs exceeded a 0.5 title-embedding cosine). The 0.78 was measured over these spurious narratives. With the clusterer corrected (no forced assignment; time-windowed clusters) the neural effect disappears while the engagement result above holds. We hand-labelled a random sample of 60 cross-source cluster pairs by reading the paired titles (labels released): 58% are the same news event, rising to 80% for pairs above 0.65 cosine. This residual cluster noise only *attenuates* measured predictability, a spurious merge injects a false transfer label uncorrelated with the narrative's real engagement, so the engagement AUC of 0.82 is a lower bound on the true signal. The synthetic results are unaffected, as narrative identity there is ground truth by construction. We

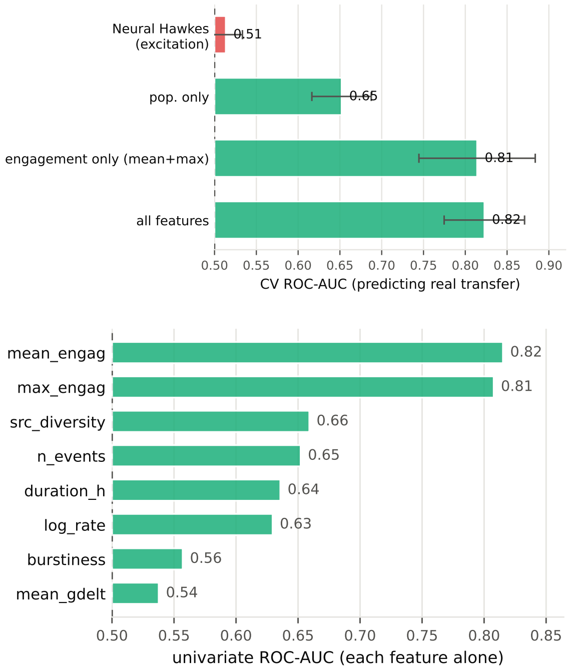

Figure 3: **What predicts real cross-platform transfer.** *Top:* 5-fold CV ROC-AUC. The neural excitation score is at chance; interpretable single-platform features reach 0.82. *Bottom:* univariate ROC-AUC per feature (each alone). Early engagement is the dominant signal (≈ 0.81); raw volume and timing/burstiness features are far weaker. Transfer is attention-driven, not excitation-driven.

Table 3: **Predicting real transfer** (corrected corpus; 4,904 test narratives, 85 transfer; stratified 5-fold CV).

Predictor	CV ROC-AUC
SimCity bridge score (neural Hawkes)	0.51 (chance)
Popularity only (event count)	0.65 ± 0.04
Engagement only (mean, max)	0.81 ± 0.07
All interpretable features	$\mathbf{0.82 \pm 0.05}$

report this in full because it is the kind of failure that peer review rarely catches but reproducibility should.

9.4 Limitations and Negative Results

We report the following limitations. (0) **Only two of the six architectural components are empirically load-bearing.** The TGN core and the cross-platform Hawkes head carry every result reported here; the SEIR-Z-D compartmental layer, the HMF bridge, the smooth Deffuant dynamics, and the dynamic influence score are *specified and released as a simulation layer but not independently validated* in this paper. We flag this explicitly rather than letting the breadth of the framework imply otherwise. (i) The multi-task objective interacts with the *timing* metric: with uniform Kendall weighting the neural model underperforms the static MHP on next-event timing (AUC 0.576), recovered by up-weighting the Hawkes task. Importantly, a loss-weight sweep ($\{1, 3, 10, 30\}\times$) shows the *transfer* AUC, our headline, is essentially flat (0.63–0.66), so the central result is not an artifact of this weighting; only the (null) timing result is weight-sensitive. (ii) Our method predicts only temporally-excited transfer; on the real corpus, where transfer is popularity-driven, it is at chance, so it is not a general-purpose transfer detector (section 9.3(3),(4)). (iii) The timing margin over static MHP, while consistent across seeds, is small and not significant. (iv) Results are on a two-platform real corpus (HN + news); the third social platform (Reddit/X) remains future work pending API access. (v) Real-corpus narratives are formed by embedding clustering of titles; a hand-labelled sample of 60 cross-source pairs gives 58% same-event precision (80% above 0.65 cosine), with the residual noise attenuating rather than inflating our predictability estimates. This limitation is also the source of our cautionary result: an earlier greedy clusterer with a forced-assignment defect inflated the real transfer AUC to 0.78 before we caught it by sampling our own clusters.

9.5 Reproducibility

All experiments run on CPU. The repository releases the generator, collectors, training (`train.py` with seed and task-weight environment controls), and one-command evaluations (`multiseed_stats.py`, `narrative_transfer_eval.py`, `platform_prediction_eval.py`, `burst_ranking_eval.py`, oracle checks), with all tables regenerable from `results/`.

References

Marián Boguñá and Romualdo Pastor-Satorras. 2002. Epidemic spreading in correlated complex networks. *Physical Review E*, 66:047104.

Pierre Brémaud and Laurent Massoulié. 1996. Stability of nonlinear Hawkes processes. *The Annals of Probability*, 24(3):1563–1588.

Qi Cao, Huawei Shen, Keting Cen, Wentao Ouyang, and Xueqi Cheng. 2017. DeepHawkes: Bridging the gap between prediction and understanding of information cascades. In *Proceedings of the 2017 ACM*

Conference on Information and Knowledge Management.

Claudio Castellano, Santo Fortunato, and Vittorio Loreto. 2009. Statistical physics of social dynamics. *Reviews of Modern Physics*, 81:591–646.

Justin Cheng, Lada Adamic, P. Alex Dow, Jon Michael Kleinberg, and Jure Leskovec. 2014. Can cascades be predicted? In *Proceedings of the 23rd International Conference on World Wide Web*.

Guillaume Deffuant, David Neau, Frederic Amblard, and Gérard Weisbuch. 2000. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3:87–98.

Nan Du, Hanjun Dai, Rakshit Trivedi, Utkarsh Upadhyay, Manuel Gomez-Rodriguez, and Le Song. 2016. Recurrent marked temporal point processes: Embedding event history to vector. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Mehrdad Farajtabar, Yichen Wang, Manuel Gomez-Rodriguez, Shuang Li, Hongyuan Zha, and Le Song. 2017. COEVOLVE: A joint point process model for information diffusion and network evolution. *Journal of Machine Learning Research*, 18:1–49.

Manuel Gomez-Rodriguez, David Balduzzi, and Bernhard Schölkopf. 2011. Uncovering the temporal dynamics of diffusion networks. In *Proceedings of the 28th International Conference on Machine Learning*.

Alan G. Hawkes. 1971. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90.

Rainer Hegselmann and Ulrich Krause. 2002. Opinion dynamics and bounded confidence: Models, analysis and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3).

Manoel Horta Ribeiro, Shagun Jhaver, Savvas Zannettou, Jeremy Blackburn, Gianluca Stringhini, Emiliano De Cristofaro, and Robert West. 2021. Do platform migrations compromise content moderation? Evidence from r/The_Donald and r/Incels. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW2):1–24.

Alex Kendall, Yarin Gal, and Roberto Cipolla. 2018. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Ryota Kobayashi and Renaud Lambiotte. 2016. TiDeH: Time-dependent Hawkes process for predicting retweet dynamics. In *Proceedings of the Tenth International AAAI Conference on Web and Social Media*.

Srijan Kumar, Xikun Zhang, and Jure Leskovec. 2019. Predicting dynamic embedding trajectory in temporal interaction networks. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Cheng Li, Jiaqi Ma, Xiaoxiao Guo, and Qiaozhu Mei. 2017. DeepCas: An end-to-end predictor of information cascades. In *Proceedings of the 26th International Conference on World Wide Web*.

Hongyuan Mei and Jason Eisner. 2017. The neural Hawkes process: A neurally self-modulating multivariate point process. In *Advances in Neural Information Processing Systems*.

Seth A. Myers, Chenguang Zhu, and Jure Leskovec. 2012. Information diffusion and external influence in networks. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Romualdo Pastor-Satorras, Claudio Castellano, Piet Van Mieghem, and Alessandro Vespignani. 2015. Epidemic processes in complex networks. *Reviews of Modern Physics*, 87:925–979.

Romualdo Pastor-Satorras and Alessandro Vespignani. 2001. Epidemic spreading in scale-free networks. *Physical Review Letters*, 86(14):3200–3203.

Marian-Andrei Rizoiu, Lexing Xie, Scott Sanner, Manuel Cebrian, Honglin Yu, and Pascal Van Hentenryck. 2017. Expecting to be HIP: Hawkes intensity processes for social media popularity. In *Proceedings of the 26th International Conference on World Wide Web*.

Emanuele Rossi, Ben Chamberlain, Fabrizio Frasca, Davide Eynard, Federico Monti, and Michael Bronstein. 2020. Temporal graph networks for deep learning on dynamic graphs. In *ICML Workshop on Graph Representation Learning*.

Polina Rozenshtein and Aristides Gionis. 2016. Temporal PageRank. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*.

Huawei Shen, Dashun Wang, Chaoming Song, and Albert-László Barabási. 2014. Modeling and predicting popularity dynamics via reinforced Poisson processes. In *Proceedings of the Twenty-Eighth AAAI Conference on Artificial Intelligence*.

Rakshit Trivedi, Mehrdad Farajtabar, Prasenjeet Biswal, and Hongyuan Zha. 2019. DyRep: Learning representations over dynamic graphs. In *International Conference on Learning Representations*.

Soroush Vosoughi, Deb Roy, and Sinan Aral. 2018. The spread of true and false news online. *Science*, 359(6380):1146–1151.

Lilian Weng, Filippo Menczer, and Yong-Yeol Ahn. 2013. Virality prediction and community structure in social networks. *Scientific Reports*, 3:2522.

Tom Wilson and Kate Starbird. 2020. Cross-platform disinformation campaigns: Lessons learned and next steps. *Harvard Kennedy School Misinformation Review*, 1(1).

Da Xu, Chuanwei Ruan, Evren Korpeoglu, Sushant Kumar, and Kannan Achan. 2020. Inductive representation learning on temporal graphs. In *International Conference on Learning Representations*.

Savvas Zannettou, Tristan Caulfield, Emiliano De Cristofaro, Nicolas Kourtellis, Ilias Leontiadis, Michael Sirivianos, Gianluca Stringhini, and Jeremy Blackburn. 2017. The web centipede: Understanding how web communities influence each other through the lens of mainstream and alternative news sources. In *Proceedings of the 2017 Internet Measurement Conference*.

Qingyuan Zhao, Murat A. Erdogdu, Hera Y. He, Anand Rajaraman, and Jure Leskovec. 2015. SEISMIC: A self-exciting point process model for predicting tweet popularity. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Simiao Zuo, Haoming Jiang, Zichong Li, Tuo Zhao, and Hongyuan Zha. 2020. Transformer Hawkes process. In *Proceedings of the 37th International Conference on Machine Learning*.

Appendix A: Notation Reference

Symbol	Meaning
$\mathbf{n}(t)$	State vector $(n_S, n_E, n_I, n_R, n_Z, n_D)$
$\mathbf{r}_k$	Jump vector for transition k
$\beta(t)$	Macroscopic SEIR-Z exposure rate
$\hat{\beta}_{v,c}(t)$	TGN node-level predicted infectivity
$\hat{\alpha}_{ij}(t)$	Neural Hawkes infectivity (platform $j \to i$)
$\hat{\gamma}_{ij}(t)$	Neural Hawkes decay parameter
γ_I	SEIR Infected-to-Recovered decay rate
σ	SEIR Exposed-to-Infected activation rate
η	Deffuant opinion convergence rate
$\zeta(t)$	Zombie resurgence rate (Hawkes-coupled)
δ_D	Archival rate ($Z \to D$)
θ	Algorithmic boost factor for Zombie content
$\epsilon_p(v)$	Platform-specific confidence bound
ρ	Deffuant smoothing steepness
τ	Platform pooling temperature
$P(k'\|k)$	Degree-degree conditional distribution
$k_v(t)$	Effective temporal degree (eq. (22))
$I(v, t)$	Dynamic influence score
$\mathbf{h}_{\mathcal{P}_i}(t)$	Platform-level pooled embedding
$\mu_i(t)$	Hawkes exogenous background rate (GDELT)
$\|\mathcal{P}\|$	Number of platforms
$\mathcal{P}_i$	Set of nodes on platform i

Appendix B: Deffuant ρ Sensitivity Analysis

The smooth formulation (eq. (25)) recovers the hard-threshold model as $\rho \to \infty$; we conjecture cluster invariance for $\rho > 10^2$, with a systematic sweep left as future work. No result in section 9.3 depends on ρ.